```verilog
`timescale 1ns / 1ps
`default_nettype none
`include "parameters.vh"
`include "control_unit.v"

module control_unit_tb;

// Testbench Signals
reg clk;
reg rst_n;
reg [31:0] i_Instr;
wire o_RegSrcE;
wire o_Sel1E;
wire o_Sel2E;
wire [3:0] o_alu_ctrl;
wire o_LoadE;
wire o_BranchE;
wire o_JalE;
wire o_JalrE;
wire o_MemSrcE;
wire o_AMOEnE;
wire [1:0] o_ResultSrcE;

// Instantiate the control unit
control_unit uut (
    .clk(clk),
    .rst_n(rst_n),
    .i_Instr(i_Instr),
    .o_RegSrcE(o_RegSrcE),
    .o_Sel1E(o_Sel1E),
    .o_Sel2E(o_Sel2E),
    .o_alu_ctrl(o_alu_ctrl),
    .o_LoadE(o_LoadE),
    .o_BranchE(o_BranchE),
    .o_JalE(o_JalE),
    .o_JalrE(o_JalrE),
    .o_MemSrcE(o_MemSrcE),
    .o_AMOEnE(o_AMOEnE),
    .o_ResultSrcE(o_ResultSrcE)
);

// Clock generation
always #5 clk = ~clk; // 10ns period (100MHz clock)

initial begin
    // Initialize
```

```verilog
    clk = 0;
    rst_n = 0;
    i_Instr = 0;

    // Reset sequence
    #10 rst_n = 1;

    // Test R-type instruction (Opcode = `R`)
    #10 i_Instr = {25'b0, `R};
    #10 $display("Testing R-type: RegSrcE=%b, Sel1E=%b, Sel2E=%b, ALUCtrl=%b",
o_RegSrcE, o_Sel1E, o_Sel2E, o_alu_ctrl);

    // Test Load instruction (Opcode = `LD`)
    #10 i_Instr = {25'b0, `LD};
    #10 $display("Testing Load: LoadE=%b, ResultSrcE=%b", o_LoadE, o_ResultSrcE);

    // Test Store instruction (Opcode = `S`)
    #10 i_Instr = {25'b0, `S};
    #10 $display("Testing Store: MemSrcE=%b", o_MemSrcE);

    // Test Branch instruction (Opcode = `B`)
    #10 i_Instr = {25'b0, `B};
    #10 $display("Testing Branch: BranchE=%b", o_BranchE);

    // Test JAL instruction (Opcode = `J`)
    #10 i_Instr = {25'b0, `J};
    #10 $display("Testing JAL: JalE=%b, ResultSrcE=%b", o_JalE, o_ResultSrcE);

    // Test JALR instruction (Opcode = `JR`)
    #10 i_Instr = {25'b0, `JR};
    #10 $display("Testing JALR: JalrE=%b, ResultSrcE=%b", o_JalrE, o_ResultSrcE);

    // Test Atomic Memory Operations (AMO) (Opcode = `AMO`)
    #10 i_Instr = {25'b0, `AMO};
    #10 $display("Testing AMO: AMOEnE=%b, ResultSrcE=%b", o_AMOEnE, o_ResultSrcE);

    // End simulation
    #20;
    $display("Testbench completed.");
    $stop;
end

endmodule
```